

NIORO DU SAHEL, Mali

WMO: 612300

Lat: 15.239N

Lon: 9.577W

Elev: 778

StdP: 14.29

Time Zone: 3.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	54.6	57.3	9.1	9.0	79.1	14.3	11.7	80.6	17.1	78.7	15.1	78.7	1.5	80	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	21.5	110.1	70.2	108.5	70.5	106.8	70.7	81.6	92.8	80.5	91.4	79.7	90.6	7.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	155.1	86.6	77.9	149.3	85.7	77.1	145.1	85.3	45.9	93.1	44.7	91.2	43.9	90.7	88.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	15.1	13.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	39.1	84.9	2.1	1.7	37.6	86.2	36.3	87.2	35.1	88.1	33.6	89.4	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	85.3	73.1	78.4	85.5	91.6	96.4	94.8	87.5	83.9	85.3	88.4	82.6	76.1
	DBStd	8.04	4.46	5.67	5.41	4.26	3.45	4.39	3.84	3.11	3.35	2.72	4.02	5.30
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	4	4	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12896	716	796	1102	1248	1437	1344	1164	1052	1060	1190	978	809
	CDD65	7425	255	376	637	798	972	894	699	587	610	725	528	344
	CDH74	107154	2897	4699	9312	13109	16769	14814	9728	7019	7619	10265	6923	4000
	CDH80	66793	1287	2607	5907	9093	12355	10541	5552	3244	3885	6225	4082	2015
Wind	WSAvg	5.6	4.8	5.9	5.8	5.7	6.1	8.0	7.0	5.8	5.0	4.6	3.9	4.7
Precipitation	PrecAvg	13.6	0.0	0.0	0.0	0.1	0.2	0.9	3.3	5.5	3.0	0.6	0.0	0.0
	PrecMax	20.4	0.1	0.4	0.4	0.5	0.9	2.1	6.4	8.9	5.7	2.5	0.3	0.3
	PrecMin	7.7	0.0	0.0	0.0	0.0	0.0	0.2	1.7	1.9	1.2	0.0	0.0	0.0
	PrecStd	2.9	0.0	0.1	0.1	0.1	0.2	0.6	1.1	1.5	1.1	0.5	0.1	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.8	103.9	107.3	110.9	112.2	111.7	104.0	98.5	101.9	104.3	102.1	98.5
		MCWB	62.1	67.4	66.8	69.6	69.0	68.9	75.2	78.2	75.1	69.2	64.8	63.4
	2%	DB	92.5	99.8	104.8	108.5	110.3	109.6	100.7	95.5	98.0	102.7	100.3	95.9
		MCWB	60.8	65.7	67.0	68.8	70.4	70.5	76.5	78.5	78.1	70.0	64.5	63.4
	5%	DB	90.1	97.1	102.6	106.7	108.9	107.5	98.5	93.2	95.7	101.2	98.6	93.7
		MCWB	59.5	65.3	66.0	68.9	71.2	71.5	76.8	78.6	78.6	70.5	64.5	62.6
	10%	DB	87.4	94.1	100.2	105.0	107.3	105.0	96.4	91.3	93.7	99.3	96.6	90.8
		MCWB	58.3	63.2	65.2	68.4	71.6	72.6	77.0	78.5	78.7	71.6	64.8	60.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.7	77.0	77.1	78.4	84.6	79.9	82.9	82.7	82.0	81.3	75.4	69.9
		MCDB	88.7	95.1	97.7	104.6	103.8	97.0	92.8	90.6	93.0	93.0	90.0	93.3
	2%	WB	65.7	72.9	73.8	75.9	80.0	78.6	80.1	81.2	80.9	79.2	71.9	67.0
		MCDB	84.8	92.0	96.0	101.4	100.7	96.5	92.1	89.7	91.5	90.9	86.0	91.6
	5%	WB	62.8	68.7	70.7	73.8	77.3	77.5	79.1	80.2	80.2	78.2	68.6	64.3
		MCDB	83.8	89.4	94.0	100.6	98.1	95.4	91.3	88.8	90.4	90.4	91.7	88.6
	10%	WB	60.1	65.3	67.8	71.6	75.9	76.5	78.3	79.4	79.6	77.2	66.3	62.2
		MCDB	83.2	90.3	92.5	98.8	97.2	94.6	90.6	88.0	89.5	89.8	91.4	86.2
Mean Daily Temperature Range	5% DB	MDBR	28.6	28.4	28.0	26.6	21.5	18.6	15.9	14.1	16.7	23.3	28.8	28.6
		MCDBR	31.7	30.8	29.1	27.4	23.5	20.8	17.6	16.1	19.3	25.9	30.2	31.0
		MCWBR	13.2	12.8	11.7	10.9	8.8	6.7	4.7	5.0	5.9	7.7	10.0	12.1
	5% WB	MCDBR	29.7	28.2	23.5	25.3	21.8	18.4	16.0	14.4	17.0	20.1	27.3	27.3
		MCWBR	16.0	15.6	11.9	13.8	9.6	5.9	6.1	6.4	6.4	7.0	12.1	12.8
Clear-Sky Solar Irradiance	taub		0.505	0.534	0.635	0.647	0.691	0.731	0.670	0.595	0.579	0.574	0.490	0.474
	taud		1.864	1.795	1.595	1.620	1.554	1.542	1.685	1.858	1.875	1.841	1.952	1.964
	Ebn at Noon		248	247	227	225	212	201	214	234	238	234	251	253
	Edh at Noon		62	69	87	85	89	89	78	66	65	65	56	55
All-Sky Solar Radiation	RadAvg		1664	1867	2083	2194	2108	2008	1890	1853	1907	1863	1714	1609
	RadStd		102	91	94	91	96	55	76	71	79	44	72	62

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page